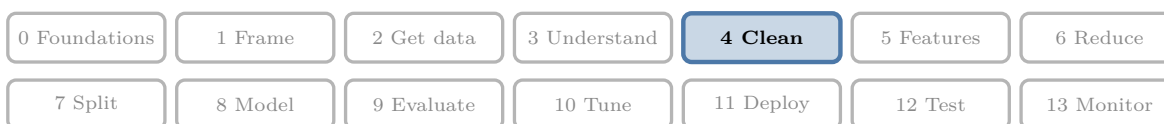


Outlier Detection with the IQR Method

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Where this fits: Pipeline map step 4 (Clean). See the [Course map](#).



- **Builds on:** Descriptive statistics ([Note 19](#)); Univariate analysis ([Note 20](#)); Outliers ([Note 41](#)); Percentile outlier method ([Note 41](#)).
- **Compare with:** Percentile outlier method ([Note 41](#)); Z-score outlier method ([Note 42](#)).

1. Overview

Key point: For a skewed column, every value beyond the box-plot fences ($Q_1 - 1.5 \text{ IQR}$ and $Q_3 + 1.5 \text{ IQR}$) is an outlier; we then trim those rows or cap the values at the fences.

Note 41 listed three rules for detecting outliers, and Note 42 put the first one, the z-score method, to work on a normal column. This Note covers the second rule: the **IQR method**, also called the **IQR proximity rule**. It is the rule to use when a column is skewed.

Figure 1 shows the whole method. We check that the column is skewed, compute the two fences from the quartiles, and then treat the values outside them by trimming or capping (both defined in Note 41, Section 7).

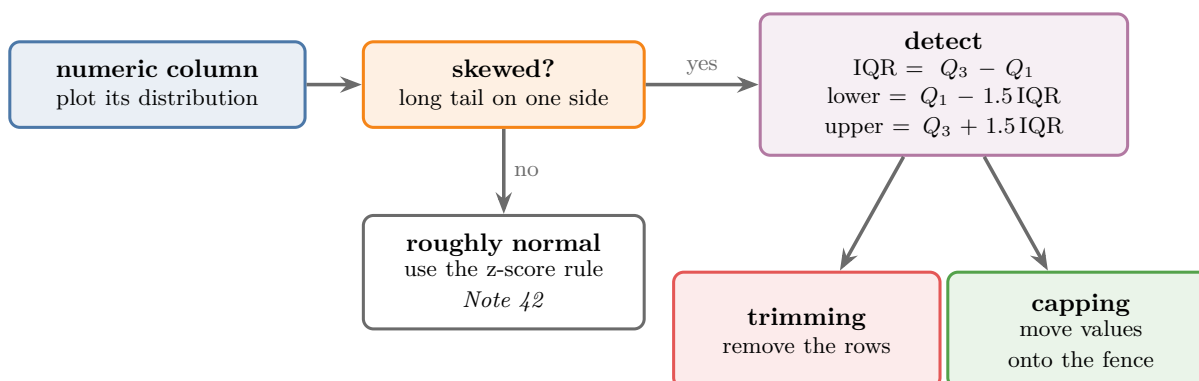


Figure 1: The IQR method: check the column is skewed, compute the fences, then trim or cap

2. The condition: a skewed column

Key point: The IQR method is meant for a column that is skewed, not normal.

A column is **skewed** when its values have a long tail on one side (Note 20). The z-score method of Note 42 does not fit such a column, because it assumes a bell shape.

The IQR method makes no such assumption. The IQR method is built only on percentiles,

which do not care about the shape of the column. To use it we need two ideas from Note 20: the box plot and the IQR.

3. The fences

Key point: The lower fence is $Q_1 - 1.5 \times \text{IQR}$ and the upper fence is $Q_3 + 1.5 \times \text{IQR}$; values outside them are outliers.

The quartiles Q_1 (25th percentile) and Q_3 (75th percentile) come from the [understanding your data Note](#) (section 7.2). The interquartile range $\text{IQR} = Q_3 - Q_1$ and the two box-plot fences come from the [univariate analysis Note](#) (sections 8 and 8.1). The IQR method uses exactly those fences as its lower and upper limits.

For the placement exam marks of Section 5, $Q_1 = 17$ and $Q_3 = 44$, so $\text{IQR} = 44 - 17 = 27$ and $1.5 \times 27 = 40.5$:

$$\text{lower} = 17 - 40.5 = -23.5, \quad \text{upper} = 44 + 40.5 = 84.5$$

A mark below -23.5 or above 84.5 is an outlier. The plan for any skewed column is therefore short: compute Q_1 , Q_3 and the IQR, compute the two fences, and trim or cap every value outside them.

Extra: The fences are robust. Q_1 and Q_3 depend only on the order of the middle values, so a few extreme values cannot drag them out, unlike the mean and standard deviation of Note 42. The factor 1.5 comes from John Tukey, who introduced the box plot (Tukey 1977). Some people also use 3: a point beyond $Q_3 + 3\text{IQR}$ (or below $Q_1 - 3\text{IQR}$) is called an “extreme” outlier, and one only beyond the 1.5 fences a “mild” one (NIST 7.1.6).

4. Treating the outliers: trimming or capping

Key point: Trimming deletes the outlier rows; capping replaces each outlier with the fence it crossed.

Both treatments come from the [outliers Note](#) (section seven, ways to treat outliers); here the fences are the limits.

5. The placement data

Key point: The data has 1,000 students; their placement exam marks are right-skewed (skewness 0.84), so the marks column is the one for the IQR method.

The data is the placement data of Note 42: one row per student of a college, 1,000 students, three columns.

cgpa	placement_exam_marks	placed
7.19	26	1
7.46	38	1
7.54	40	1
6.42	8	1

- **cgpa**: the student's CGPA (out of 10).
- **placement_exam_marks**: marks out of 100 in the aptitude test held before placement.
- **placed**: 1 if the student got a job offer, 0 if not.

Note 42 (Figure 3) showed the shapes: **cgpa** is a bell (skewness -0.01), while **placement_exam_marks** has a long tail to the right (skewness 0.84). So **placement_exam_marks** is the candidate for the IQR method.

The summary numbers of the marks column tell the same story:

Count	Mean	Std	Min	25%	50%	75%	Max
1,000	32.23	19.13	0	17	28	44	100

A quarter of the students scored below 17, half below 28 and three quarters below 44. One student scored 0 and one scored the full 100, which suggests a hard exam with a few toppers.

Python: Loading the data and checking the marks column.

```
import pandas as pd

df = pd.read_csv("data/placement.csv")
df.shape      # (1000, 3)

df["placement_exam_marks"].skew()      # 0.84
df["placement_exam_marks"].describe()  # the table above
```

6. Finding the fences and the outliers

Key point: $Q_1 = 17$ and $Q_3 = 44$ give an IQR of 27 and fences of -23.5 and 84.5 ; 15 students lie above the upper fence and none below the lower one.

Figure 2 shows the box plot of the marks above their histogram. The box runs from $Q_1 = 17$ to $Q_3 = 44$, so the IQR is 27, and the fences sit at -23.5 and 84.5 (Section 3).

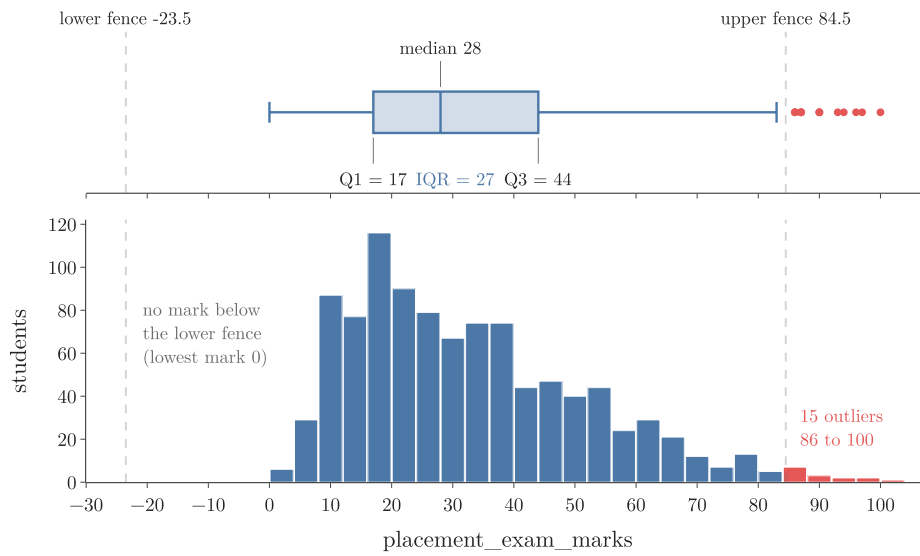


Figure 2: The placement exam marks: box plot with Q1, median and Q3, the two fences, and the 15 outliers in red

- **Upper fence, 84.5:** 15 students scored above it, with marks from 86 to 100. In the box plot they are the red dots on the right.
- **Lower fence, -23.5:** marks cannot be negative, so no student lies below it. With a right-skewed column, all the outliers sit on the long-tail side.

The 15 outliers:

Mark	Students	Rows
86	3	40, 61, 771
87	4	283, 290, 311, 685
90	3	162, 324, 730
93, 94, 96, 97, 100	1 each	134, 9, 630, 846, 917

Python: The quartiles, the IQR and the fences.

```
marks = df["placement_exam_marks"]

percentile25 = marks.quantile(0.25)    # 17.0
percentile75 = marks.quantile(0.75)    # 44.0
iqr = percentile75 - percentile25       # 27.0

upper_limit = percentile75 + 1.5 * iqr  # 84.5
lower_limit = percentile25 - 1.5 * iqr  # -23.5

df[marks > upper_limit]                 # 15 rows
df[marks < lower_limit]                 # 0 rows
```

`quantile(0.25)` returns the 25th percentile; `quantile` takes the fraction (0.25), not the percent (25).

7. Trimming in code

Key point: Keeping only the rows inside the fences removes the 15 outliers and leaves 985 rows.

Trimming is a filter: keep the rows whose mark lies between the two fences. Since no mark is below the lower fence, keeping the marks below 84.5 is enough here.

Python: Trimming.

```
# keep the rows with lower <= mark <= upper
new_df = df[marks.between(lower_limit, upper_limit)]
new_df.shape      # (985, 3)
```

Figure 3 (middle row) shows the result. The histogram barely changes: only its thin right tail is gone. The box plot loses its row of red dots.

The mean drops from 32.23 to 31.34 and the skewness from 0.84 to 0.65.

7.1 Why a new outlier appears after trimming

Key point: The box plot of the trimmed data computes new fences from the 985 remaining rows, so a value that was inside before can now be outside.

The trimmed box plot in Figure 3 still shows one red dot, at a mark of 83. It is not a mistake. A box plot always computes its fences from the data it is given.

For the 985 trimmed rows, Q_3 drops from 44 to 43, so the new upper fence is $43 + 1.5 \times (43 - 17) = 82$. The mark 83, safely inside the old fence of 84.5, is now just outside the new one.

In this Note we detect once, with the fences of the original data, and treat once. (Trimming again here would remove the 83 and then stop: a third round finds nothing. The last cell of the Notebook shows this.)

8. Capping in code

Key point: Each mark above 84.5 becomes 84.5 (and each mark below -23.5 would become -23.5); all 1,000 rows stay.

Capping goes through the column value by value, exactly as in Note 42:

- above the upper fence: replace it with the upper fence;
- below the lower fence: replace it with the lower fence;
- otherwise: leave it as it is.

NumPy's `np.where(condition, value if true, value if false)` does this for the whole column; two conditions need two calls, one inside the other.

Python: Capping with `np.where`.

```
import numpy as np
```

```
new_df_cap = df.copy()
col = new_df_cap["placement_exam_marks"]
new_df_cap["placement_exam_marks"] = np.where(
    col > upper_limit,
    upper_limit,
    np.where(col < lower_limit, lower_limit, col))
```

```
new_df_cap.shape    # (1000, 3): no row lost
```

`df.copy()` makes an independent copy, so the original `df` keeps its values for the comparison below.

The result, compared with the original and the trimmed column:

	Original	Trimmed	Capped
Rows	1,000	985	1,000
Mean	32.23	31.34	32.14
Standard deviation	19.13	17.86	18.87
Maximum	100	83	84.5
Skewness	0.84	0.65	0.76

Figure 3 (bottom row) shows the capped column. The 15 outliers now all sit at 84.5, so the histogram rises in that one spot (the orange bar). The box plot has no dots: its whisker ends exactly at the fence.

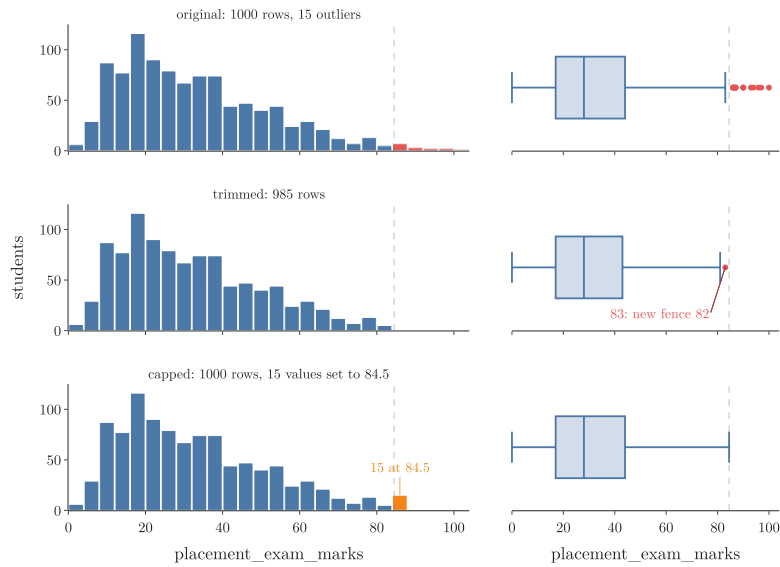


Figure 3: Before and after: the original column, the trimmed column (985 rows) and the capped column (1,000 rows, outliers moved onto 84.5); histograms on the left, box plots on the right, the upper fence dashed

Extra: pandas has a one-line shortcut for capping, `clip`. It gives exactly the same column as the two nested `np.where` calls.
`marks.clip(lower_limit, upper_limit)`

9. Learning the fences on the training set

Key point: The fences are learned from the data, so they should be learned on the training set only and then applied to the test set.

The fences follow the same train-only rule as the z-score limits of the [z-score Note](#) (section 11, learning the limits on the training set).

Extra: With an 80/20 split (`random_state=42`), the 800 training rows have the same $Q_1 = 17$ and $Q_3 = 44$, so the fences stay at -23.5 and 84.5 :

	Rows	Outliers (training fences)
Training set	800	14: marks 86 to 97
Test set	200	1: mark 100

Quartiles move very little between 800 and 1,000 rows, so here the result is the same. Trimming the training set leaves 786 rows; the Notebook shows both steps.

10. Strengths and limits of the IQR method

Key point: The method is simple, works on skewed columns and is not pulled by the outliers it looks for; it is only a rule of thumb.

- **Simple:** two percentiles give both fences.
- **Shape-free:** it needs no bell shape, so it fits skewed columns.
- **Robust:** the quartiles are not pulled by extreme values (Section 3, Extra).

Extra: Two things to keep in mind.

- **On a long tail, real values get flagged.** In a strongly skewed column the far tail can be perfectly genuine (a few very high incomes, a few toppers; here, 15 real exam marks between 86 and 100). The fences flag it all the same. So we still decide, as in Note 41 (Section 4), whether those values are errors or real.
- **One side may never be used.** For a right-skewed column with a natural lower bound, such as marks starting at 0, the lower fence can lie below every possible value, as it does here (-23.5). That is expected, not a bug.

11. Summary

Step	What we do	On the placement marks
Check the shape	plot the column; the method is for skewed columns	right-skewed, skewness 0.84
Quartiles	$Q_1 = 25\text{th}$, $Q_3 = 75\text{th}$ percentile, $\text{IQR} = Q_3 - Q_1$	$Q_1 = 17$, $Q_3 = 44$, $\text{IQR} = 27$
Detect	fences $Q_1 - 1.5 \text{ IQR}$ and $Q_3 + 1.5 \text{ IQR}$	-23.5 and 84.5 ; 15 outliers, all above
Trim	keep the rows inside the fences	985 rows left
Cap	move values beyond a fence onto it	1,000 rows; maximum 84.5
Better practice	learn the fences on the training set	same fences; 14 outliers in train, 1 in test

	Z-score method (Note 42)	IQR method (this Note)
Column shape	roughly normal	skewed
Built on	mean and standard deviation	Q_1 and Q_3
Limits	$\mu \pm 3\sigma$	$Q_1 - 1.5 \text{ IQR}$, $Q_3 + 1.5 \text{ IQR}$
Pulled by outliers	yes	hardly

- The IQR method is for skewed columns; the z-score method of Note 42 is for normal ones.
- The IQR is $Q_3 - Q_1$, the width of a box plot's box.
- The fences are $Q_1 - 1.5 \text{ IQR}$ and $Q_3 + 1.5 \text{ IQR}$: the same limits a box plot uses for its dots.
- Values outside the fences are outliers; we trim them (drop the rows) or cap them (set them to the fence).
- A box plot of trimmed data computes new fences, so a new dot can appear; we do not trim again.
- The fences should be learned on the training set only.

12. Sources

- Tukey, J. W. (1977). *Exploratory Data Analysis*. Addison-Wesley.
- NIST/SEMATECH. *e-Handbook of Statistical Methods*, Section 7.1.6, "What are outliers in the data?". <https://www.itl.nist.gov/div898/handbook/>

13. Key terms

Term	Meaning
IQR method (IQR proximity rule)	Outlier detection that flags values beyond 1.5 IQR outside the box (Q_1 to Q_3); for skewed columns
<code>quantile</code>	pandas method that returns a percentile, given as a fraction (0.25 for the 25th)